Modeling Credit Risk

Developing a Macro-Conditioned Probability of Default Transition Matrix

Project Overview

- Credit rating migrations forecasted using Markov Chains
- 1-quarter (3-month) time horizon for high-resolution tracking
- Quantifies likelihood of rating migrations and default

Methodology & Tech Stack

BUILT WITH PYTHON & MACRO DATA

Python Ecosystem

- Built entirely in Python
- **pandas** used for data wrangling and matrix manipulation
- **statsmodels** applied for core econometric equations

Macro-Conditioning

- Probabilities dynamically conditioned on macroeconomic variables
- Incorporates GDP growth, unemployment rate, treasury yields
- Reflects shifts in the broader economic cycle

Rigorous Selection

- Data-driven variable selection process
- Utilizes stationarity checks and p-value significance
- Predictive power analysis ensures robust indicators

The 1-Quarter Transition Matrix

MACRO-CONDITIONED OUTPUT DATA (%)

INITIAL \ FINAL	AAA	AA	A	BBB	BB	B	CCC	DEFAULT
AAA	98.50	1.30	0.15	0.03	0.02	0.00	0.00	0.00
AA	0.20	97.80	1.70	0.20	0.05	0.03	0.02	0.00
A	0.05	0.60	97.50	1.50	0.20	0.10	0.03	0.02
BBB	0.01	0.10	2.50	94.00	2.80	0.50	0.05	0.04
BB	0.01	0.05	0.30	2.50	92.50	4.00	0.40	0.24
B	0.00	0.02	0.10	0.30	3.50	93.00	2.00	1.08
CCC	0.00	0.00	0.05	0.20	1.00	4.50	88.00	6.25

Core Mathematical Properties

MARKOV ASSUMPTIONS

The Markov Property

- Future states depend solely on current state
- Historical paths are irrelevant (memoryless property)

$$P(X_{t+1} = j \mid X_t = i) = p_{ij}$$

Diagonal Dominance

- Extremely high probabilities on matrix diagonal
- Reflects massive rating 'stickiness' over short horizon
- Majority of entities maintain stable ratings

Absorbing States

- 'Default' treated as an absorbing state
- Entities entering default remain permanently
- Probability of transitioning out of default is zero

Visualizing Migration Risk

STARTING STATE: 'BBB' RATING

Forecasted probability distribution for a 'BBB' rated borrower after 1 quarter: